

FINANCIAL NEWS SLATE

PRODUCTION TECHNICAL BLUEPRINT & DEPLOYMENT SPECIFICATION

Complete Technical Architecture, Data Schemas, Scraper Engines, Component Hierarchies, and CI/CD Pipeline
Target Project: masterchetan-financial.web.app | React 18 + Vite 8 + TailwindCSS + Firestore

1. CORE ARCHITECTURE & ZERO-QUOTA CDN PATTERN

This system employs a dual-data pipeline. High-frequency live tickers, 8 Nifty sectoral indices, and the 5-Day FII/DII Institutional Table are fetched from a static CDN JSON file (public/market-data.json). This guarantees ZERO Firestore database read quotas for high-frequency client updates.

Production Technical Stack:

- Frontend: React 18, Vite 8, TailwindCSS (Modern Light Theme, #e02020 Red Accent), Lucide-React, Framer Motion
- Data Engine: Public Static CDN Dataset (public/market-data.json) + Cloud Firestore NoSQL Database (articles)
- AI Processing: Google Gemini 1.5 Flash API (@google/genai) for 3-bullet summarization & category tagging
- Automation Pipeline: GitHub Actions Headless Cron (.github/workflows/curate.yml) twice daily (9:03 AM & 5:33 PM IST)
- Hosting CDN: Firebase Hosting CDN (masterchetan-financial.web.app)

2. COMPONENT ARCHITECTURE & WIDGET SPECIFICATIONS

Header.jsx	Brand Masthead, WhatsApp (93242 73030), Advisory Contact Cards, Saved Bookmarks Filter
MarketTicker.jsx	Live Marquee Bar for SENSEX, NIFTY 50, BANK NIFTY, GOLD (24K), USD/INR, CRUDE BRENT
SectorHeatmap.jsx	8 Nifty Sectoral Indices Grid with Direct Client Live Fetch to Yahoo Finance on Refresh
FiiDiiTable.jsx	Official 5-Day FII & DII Institutional Gross Buy, Gross Sell & Net Cash Flow Table + MTD Row
EventCalendar.jsx	Upcoming Macro Catalysts Strip & RBI MPC Policy Watch
ArticleModal.jsx	Full-Screen Article Reader Drawer Modal + Deep Link URL Parameter (?article=ID)

3. MARKET & INSTITUTIONAL DATA SCRAPER ENGINE (scripts/fetch-market-data.cjs)

```
const fs = require('fs');
const path = require('path');

// Scrapes Google Finance, Yahoo Finance & Economic Times exchange tables
async function fetchAllLiveMarketData() {
  const tickerResults = [];
  const sensex = await fetchGoogleFinanceQuote('SENSEX:INDEXBOM', 'SENSEX');
  if (sensex) tickerResults.push(sensex);

  const nifty = await fetchGoogleFinanceQuote('NIFTY_50:INDEXNSE', 'NIFTY 50');
  if (nifty) tickerResults.push(nifty);

  const fiiDiiTableData = await fetch5DayFiiDiiTable();

  const payload = {
    ticker: tickerResults,
    fiiDiiTable: fiiDiiTableData,
    updatedAt: new Date().toISOString()
  };

  fs.writeFileSync(path.join(__dirname, '../public/market-data.json'), JSON.stringify(payload, null, 2));
}
fetchAllLiveMarketData();
```

4. GEMINI AI CURATION PIPELINE (scripts/curate-news.cjs)

```
const { GoogleGenerativeAI } = require('@google/generative-ai');
const admin = require('firebase-admin');

const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY);
const model = genAI.getGenerativeModel({ model: 'gemini-1.5-flash' });

// Summarizes raw articles into 3 bullets, categorizes & assigns High/Medium/Standard impact
async function curateArticle(rawItem) {
  const prompt = `Summarize in 3 bullet points, categorize & assign impact: ${rawItem.content}`;
  const res = await model.generateContent(prompt);
  const parsed = JSON.parse(res.response.text().replace(/``json|``/g, '').trim());

  await admin.firestore().collection('articles').doc(rawItem.id).set({
    title: rawItem.title, summary: parsed.summary, category: parsed.category,
    impactLevel: parsed.impactLevel, curatedDate: new Date().toISOString().split('T')[0]
  }, { merge: true });
}
```

5. GITHUB ACTIONS AUTOMATION WORKFLOW (.github/workflows/curate.yml)

```
name: ⌚️ Daily Market & Financial Curation Pipeline

on:
  schedule:
    - cron: '33 3 * * *' # 9:03 AM IST
    - cron: '3 12 * * *' # 5:33 PM IST
  workflow_dispatch:

jobs:
  curate-and-deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - uses: actions/setup-node@v3
        with: { node-version: 18 }
      - run: npm ci
      - run: node scripts/fetch-market-data.cjs
      - run: node scripts/curate-news.cjs
      env:
        GEMINI_API_KEY: ${ secrets.GEMINI_API_KEY }
        FIREBASE_SERVICE_ACCOUNT_KEY: ${ secrets.FIREBASE_SERVICE_ACCOUNT_KEY }
      - run: npm run build
      - uses: FirebaseExtended/action-hosting-deploy@v0
        with:
          repoToken: '${ secrets.GITHUB_TOKEN }'
          firebaseServiceAccount: '${ secrets.FIREBASE_SERVICE_ACCOUNT }'
          channelId: live
          projectId: masterchetan-financial
```

6. REPLICATION & CUSTOMER ONBOARDING STEPS

1. Clone Working Repository: `git clone https://github.com/masterschetanapp-creator/masterschetan-news-slate.git new-project`
2. Firebase Target Setup: Change `projectId` in `.firebaserc` to client's Firebase project ID.
3. GitHub Repo Secrets: Configure `GEMINI_API_KEY`, `FIREBASE_SERVICE_ACCOUNT`, and `FIREBASE_SERVICE_ACCOUNT_KEY`.
4. Custom Subject Matter: Replace category lists in `CategoryFilter.jsx` and prompt definitions in `curate-news.cjs`.
5. Initial Production Build: Run `npm run build` and `firebase deploy --only hosting`.

